

Diversification of forensic data analysis tools through the application of TDA

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May 26, 2023

joint work with

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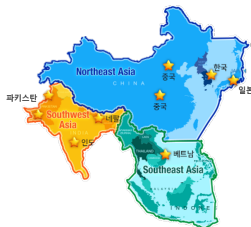
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1 AISNP data

AISNP data for Asian ancestry prediction

- Biogeographical ancestry information of unidentified individuals serves as a valuable indicator in forensic investigation.
- Asia is the largest continent with diverse ethnic groups, it is especially important to investigate AIMs for the Asian population.



1 AISNP data

Data collection



DEPARTMENT OF FORENSIC MEDICINE
서울대학교 의과대학 법의학교실

Samples to study

Geographical Region	Population		Sample size
Northeast Asia	China	[CHS]	113
		[CHB]	100
		[CDX]	102
	Japan	[JPT]	101
	Korea	[KOR]	345
Southeast Asia	Vietnam	[KHV]	100
		[VNI]	199
	Myanmar	[MYA]	199
	Bangladesh	[BEB]	104
Southwest Asia	Sri Lanka	[STU]	111
		[NP]	148
		[ITU]	111
	India	[GIH]	108
		[IN]	51
		[PT]	89
	Pakistan	[PJL]	108
		[PJ]	50
			2139

Workflow

Applied Biosystems™ Precision ID Ancestry Panel	
Targets	165 single-nucleotide polymorphisms (SNPs)
Amplicon length	Average 130 bp for the 55 SNPs from Dr. Kenneth Kidd Average 120 bp for the 123 SNPs from Dr. Michael Seldin
Primer pool size	165 primer pairs in 1 tube
Input DNA	1ng

① Construct library

Precision ID Ancestry panel



② Prepare Template

Ion Chef



③ Run Sequence



Ion S5
GeneStudio

④ Analyze data

Torrent Suite Software



from Precision ID NGS system brochure

1 AISNP data

	A	B	C	D	E	F	G
1	Geographic region	POP-ID	SNU-ID	rs2986742	rs6541030	rs647325	rs4908
2	NEA	CDX	SNU-CDX-001	TT	GG	AG	AC
3	NEA	CDX	SNU-CDX-002	CT	GG	GG	AC
4	NEA	CDX	SNU-CDX-003	CT	AG	GG	AC
5	NEA	CDX	SNU-CDX-004	CT	AG	AG	GC
6	NEA	CDX	SNU-CDX-005	TT	AG	AG	AC
7	NEA	CDX	SNU-CDX-006	TT	GG	AA	AA
8	NEA	CDX	SNU-CDX-007	TT	GG	GG	GC
9	NEA	CDX	SNU-CDX-008	TT	AG	AG	AA
10	NEA	CDX	SNU-CDX-009	CC	GG	GG	AA
11	NEA	CDX	SNU-CDX-010	TT	AG	GG	AC
12	NEA	CDX	SNU-CDX-011	CT	GG	GG	AC
13	NEA	CDX	SNU-CDX-012	TT	GG	AG	AC

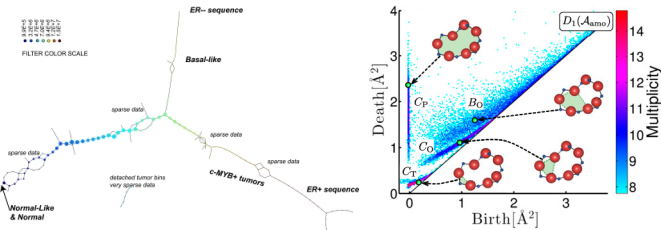
a small part of collected data

Entire dataset consists of 2,139 samples and 165 SNPs.

2 A glimpse on TDA

TDA(Topological Data Analysis)

- Roughly speaking, **topology**, a branch of mathematics, studies **shape** of spaces(figures).
- TDA aims to obtain insight by studying **shape** of data.



(left) breast cancer patients described by mapper (M. Nicolau, A. J. Levine, G. Carlsson. 2011)

(right) structure of amorphous solid in persistence diagram (Y. Hiraoka et al. 2015)

2 A glimpse on TDA

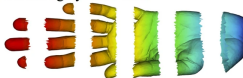
Mapper

- Mapper describes shape of data as a network.
- **Every network is embeddable in 3d space,** no matter how complicated it is.

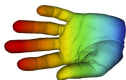
A Original Point Cloud



C Binning by filter value



B Coloring by filter value



D Clustering and network construction

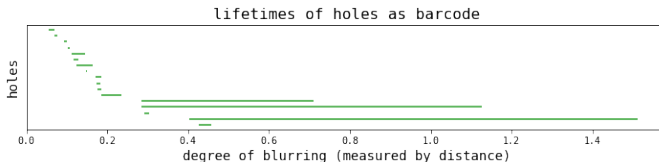
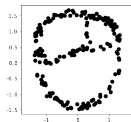


mapper at a glance (P. Y. Lum et al. 2013)

2 A glimpse on TDA

Persistence homology

- Homology counts holes of a figure,
- and scale of such hole is given in terms of persistence.
- **More persistent holes represent shape of data.**



(upper) original data points on leftmost and its blurred versions

(lower) birth-death bars of each holes

3 AISNP data and TDA

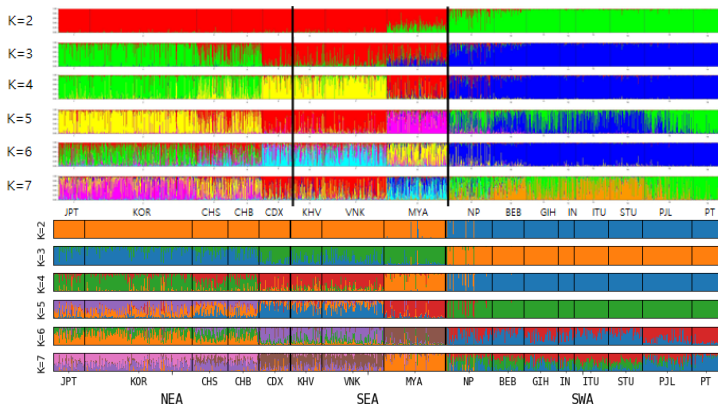
	rs123	rs45	rs678
Aaron	GG	AA	AG
Betty	AG	AA	AG
Cathy	AG	AG	GG

- Q. How can we regard each sample as a point (in space)?
- A. By giving **distance** between samples.
- Hamming distance(the proportion of distinct entries) is a possible choice for SNP data.

- Aaron $\leftrightarrow$ Betty = $1/3$, Betty $\leftrightarrow$ Cathy = $2/3$
- rs123 $\leftrightarrow$ rs45 = $2/3$, rs45 $\leftrightarrow$ rs678 = 1

3 AISNP data and TDA

Population structure investigation

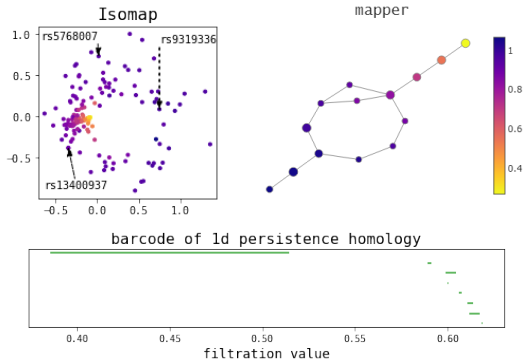


(upper) result produced by software *STRUCTURE* (Pritchard, J. K. et al. 2000)

(lower) result of fuzzy clustering

3 AISNP data and TDA

Observation on set of variables



(upper left) visualization by Isomap

(upper right) visualization by mapper

(bottom) persistent homology as barcode

The set of variables has nontrivial shape!

4 Application to machine learning

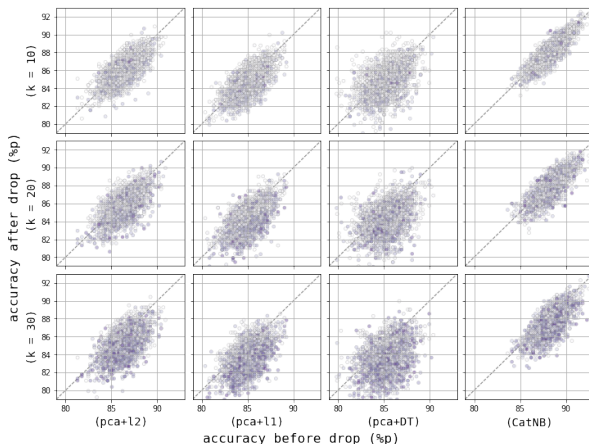
Q. Which subset of SNP variables are more useful for region classification?

A. Presumably, of which shape maintains that of full set more closely

How to verify:

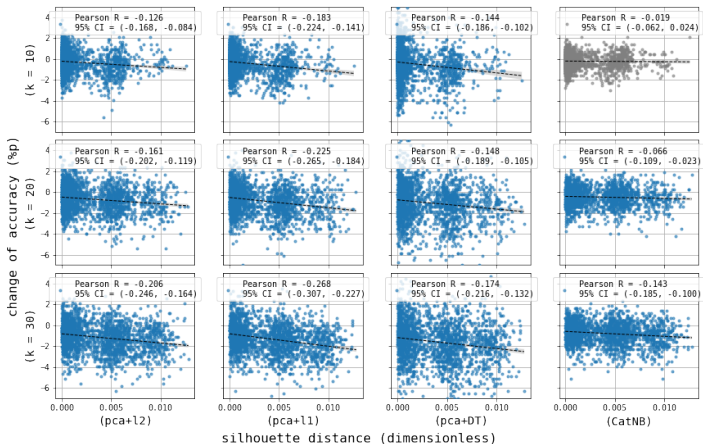
- 1 With all of 165 variables, **calculate shape** of set of loci, and **evaluate classification models**.
- 2 Drop k variables randomly, and **calculate shape**, and **evaluate models** as above.
- 3 Compare **change of shape** and **change of performance**, and repeat N times.

4 Application to machine learning



result of experiment under several distinct conditions
deeper (purple) color indicates more change in shape
 k is fixed on each row, and model on each column, all with $N = 500$

4 Application to machine learning



result of experiment under several distinct conditions

k is fixed on each row, and model on each column, all with $N = 500$

Discussion

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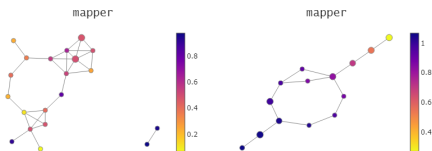
Application to
machine learning

Conclusion

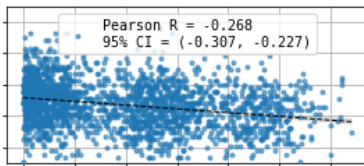
- Although being weak with large variation, correlation is statistically significant for most cases.
- Dropping variables did not result in increase in accuracy (on average). When less number of SNPs must to be used, however, there are choices with less decrease in performance.
- Specific shape (circle) itself is still of interest, and its meaning should be explored.

Take away messages

- 1 AISNP data has certain shape, as revealed by TDA.



- 2 The shape can help identifying ancestry information.



End of presentation

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Thank you!